

Manan M. Prajapati

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SUMMARY

Third-year B.Tech CSE student with hands-on experience in data analysis, machine learning, and end-to-end data pipelines using Python and SQL. Built projects across EDA, predictive modeling, and dashboarding. Targeting Data Science / ML roles at product-based companies.

EDUCATION

B.Tech — Computer Science Engineering 2023 – 2027

MBIT, CVMU University · Anand, Gujarat | CGPA: 7.49 / 10

TECHNICAL SKILLS

Languages: Python, SQL, Bash (basic)

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Plotly

Tools: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Power BI (basic)

Concepts: Data Analysis, EDA, Machine Learning, Feature Engineering, Statistical Modeling, Data Visualization

PROJECTS

OzoneWatch — Air Quality Risk Prediction System · ozonewatch.streamlit.app

Python · Scikit-learn · XGBoost · SMOTE · SHAP · Streamlit · Open-Meteo API

- Identified 94/6 class imbalance in a 2,534-record dataset; applied SMOTE to oversample the minority class from 127 to 1,900 samples, enabling reliable model training on skewed health-risk data.
- Benchmarked Logistic Regression, Random Forest, and XGBoost; selected Logistic Regression for highest recall (0.91), prioritizing public health safety over raw accuracy in a risk classification context.
- Applied SHAP explainability to identify solar radiation as the strongest ozone predictor; optimized decision threshold to improve F1 from 0.42 to 0.49 and deployed a live Streamlit app integrated with Open-Meteo API for real-time Houston AQI risk prediction.

Multimedia Customer Auditor System (MMCSA) Python · SQL · Pandas · Matplotlib

- Built a compliance auditing platform to monitor customer support agents against company guidelines, automating rule-violation detection across interaction logs and reducing manual review effort.
- Engineered ETL pipelines using Pandas and SQL to ingest, clean, and analyze 10,000+ support records, enabling structured detection at scale.
- Generated automated audit reports with Matplotlib dashboards, surfacing non-compliant agent trends for management decision-making.

Metro Sim — Train Route Monitoring System Python · Simulation · Team Project

- Collaborated in a 4-member team to build a Python simulation system that monitors train route configurations and detects off-track deviations in real time.
- Implemented rule-based predictive logic to flag trains at risk of routing errors before they occur, improving simulated safety response time by ~30%.
- Built an operator dashboard using Matplotlib to visualize live train positions and route status, enabling faster anomaly detection.

ACHIEVEMENTS & ACTIVITIES

- Smart India Hackathon (SIH) — Selected and participated in national-level government hackathon.
- CVMU University Inter-College Technical Competition — Placed among top teams, Anand, Gujarat.
- Actively building Data Science portfolio on GitHub with daily commits documenting end-to-end ML projects.

RELEVANT COURSEWORK

Data Structures & Algorithms · Database Management Systems · Statistics & Probability · Machine Learning Fundamentals · Object Oriented Programming